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(54) **SYSTEMS AND METHODS FOR GENERATING USE OF FORCE INDICATORS**

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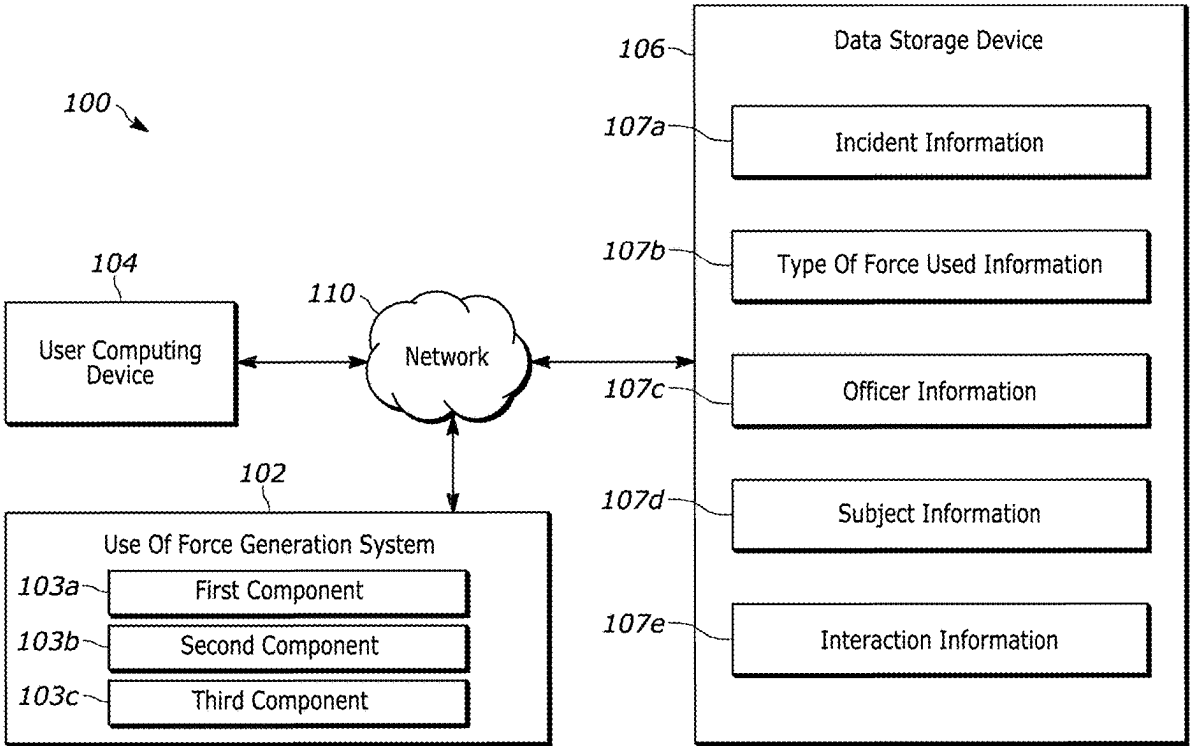
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(57) **ABSTRACT**

A system and method for determining a score reflecting the use of force that a police officer applied in an incident, comprising a computer system for collecting, storing and processing data relating to an incident involving the police officer's use of force in one or more interactions within an incident; the computer system iteratively generating scores for each use of force interactions within the incident; the computer system comparing the police officer's use of force score, whether overall or for an interaction to a score for a group of officers; and, the computer system generating a risk score by running a model based on the use of force score.



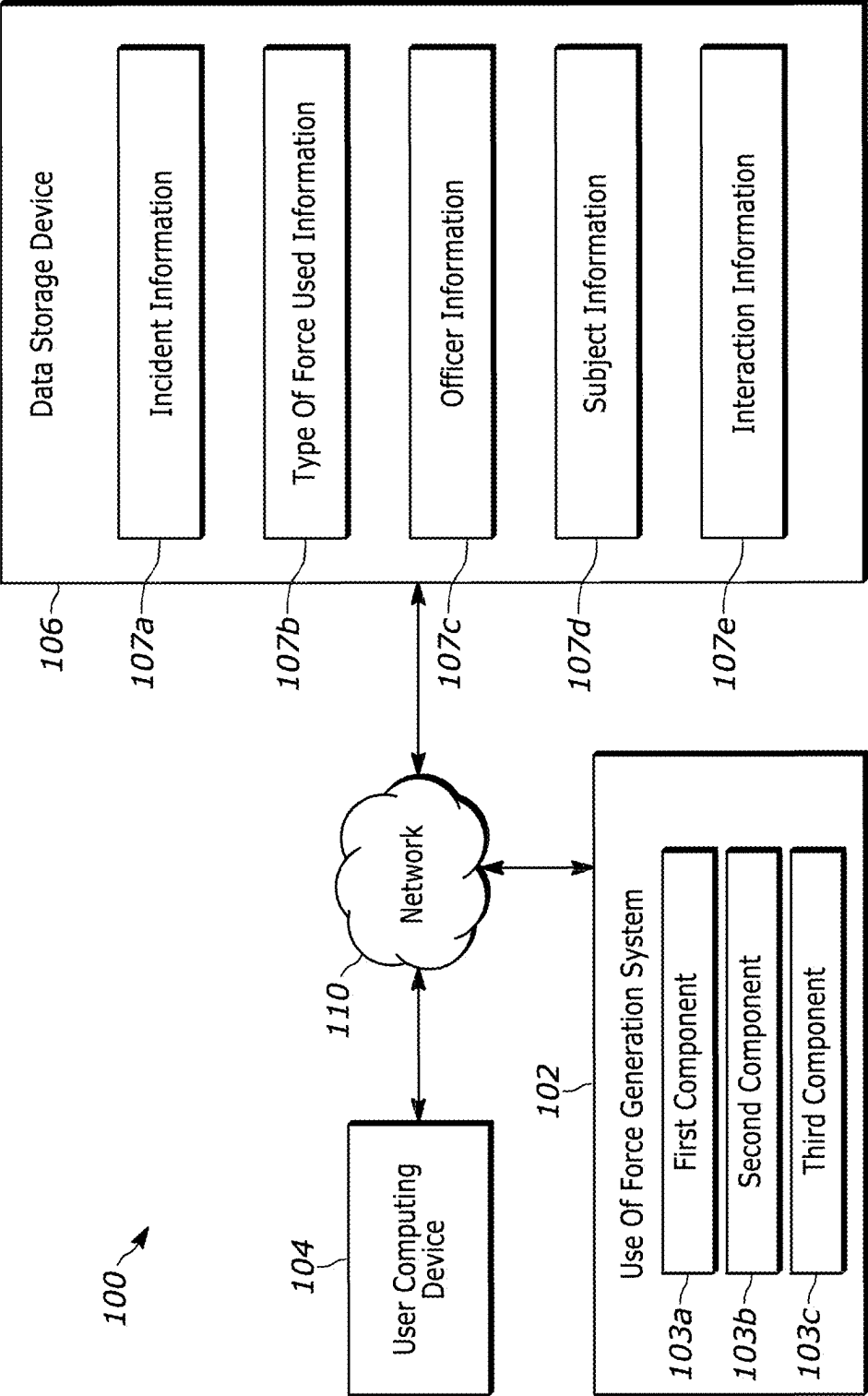
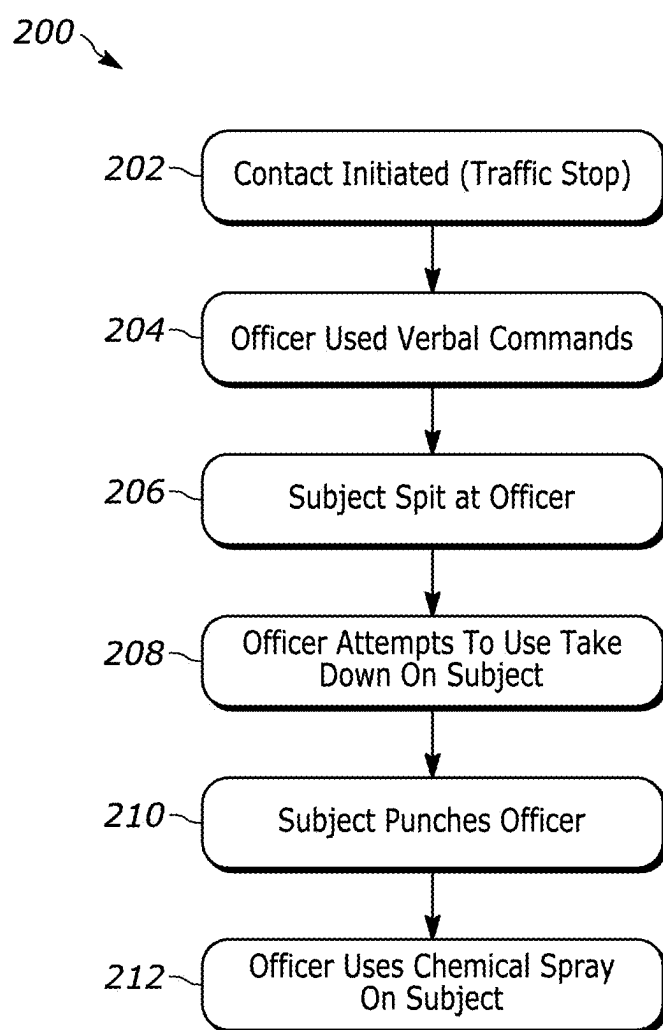


FIG. 1



Incident 1 - Interaction Sequence

FIG. 2

Initial Contact

Reason For Initial Contact*

Traffic Stop

Officer-Subject Use Of Force Sequence 1

Officer C. Smith

Used Verbal Commands

Subject John Doe

Position When Applying Action

Standing

Officer-Subject Use Of Force Sequence 2

Subject John Doe

Spit At

Officer C. Smith

Position When Applying Action

Other

In Car

FIG. 3

400

Officer-Subject Use Of Force Sequence 3

Officer C. Smith 404 Used Take Down On 406 Subject John Doe

Select Area Of Body Where Force Was Applied On Subject:

402

416 414 412 410 408

418

Body Contact Front

Right Chest X Left Chest X

Left Shoulder X Right Shoulder X

Body Contact Back Please Select

Position When Applying Action Standing 420

Protective Barrier Used No 422 Subject Struck? Yes 422

FIG. 4

500

504

506

Officer-Subject Use Of Force Sequence 4

Subject John Doe

Used Fists

Officer C. Smith

502

Select Area Of Body Where Force Was Applied On Subject:

510

508

512

Body Contact Front

Right Cheek X

Body Contact Back

Please Select

Position When Applying Action

Other

514

In Car

514

FIG. 5

600

Officer-Subject Use Of Force Sequence 5

604

606

Officer C. Smith

Discharged Chemical At

Subject John Doe

Select Area Of Body Where Force Was Applied On Subject:

602

610

612

614

608

616

Body Contact Front

Right Eye X

Left Eye X

Nose X

Body Contact Back

Please Select

Chemical Type

Chemical Serial Number

618

Position When Using Chemical

Please Select

626

Protective Barrier Used

No

628

Subject Struck?

Yes

620

630

Subject Decontaminated After Usage?

No

632

Accidental Discharge?

No

634

622

Distance When First Used

Next To Driver Side Window

624

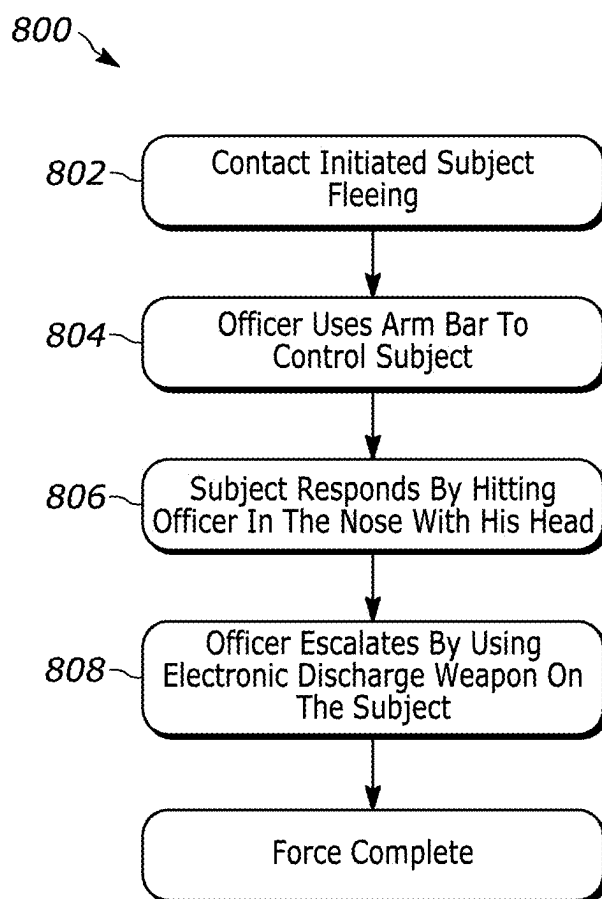
Distance When Last Used

FIG. 6

700 

Subject Action	Officer Action	Subject Action Score	Officer Action Score
Traffic Violation	Verbal Commands	0	1
Subject Spit	Physical-Take Down	1	2
Subject Punch	Chemical Spray	2	3
	Total	3	6
	Score		
Starting Subject Action	0		
Starting Officer Action	1		
Starting Proportionality	0		
Starting Escalation	1		
Total Subject Action	3		
Total Officer Action	6		
Subject Highest Action	2		
Officer Highest Action	3		
Highest Action Proportion	0		
Highest Proportionality	0		
Subject Escalation	1		
Officer Escalation	3		

FIG. 7



Incident 2 - Interaction Sequence

FIG. 8

900

904

906

Officer Use Of Force 1

Officer A. Miller

Used Arm Bar On

Subject Person

Select Area Of Body Where Force Was Applied On Subject:

902

910

908

FIG. 9

1000

1004

1006

Officer Use Of Force 2

Subject Person

Used Head

Officer A. Miller

Select Area Of Body Where Force Was Applied On Subject:

1002

1010

1008

Body Contact Front

Nose X

Body Contact Back

Please Select

Missed Attempted Contact With Subject

No

1012

1014

FIG. 10

1100

1104

1106

Officer Use Of Force 3

Officer A. Miller

Used CED On

Subject Person

Select Area Of Body Where Force Was Applied On Subject:

1102

1108

1110

Body Contact Front

Please Select

Body Contact Back

Lower Back X

1112

Missed Attempted Contact With Subject

No

1114

Done With Actions

+ Add More Actions

The figure shows a software window titled "Officer Use Of Force 3". At the top, there are three dropdown menus: "Officer A. Miller", "Used CED On", and "Subject Person". Below these is a section titled "Select Area Of Body Where Force Was Applied On Subject:". This section contains two human body diagrams. The left diagram is a front view, and the right diagram is a back view. The back view has a shaded area on the lower back, labeled "Lower Back X". Below the front view is a button labeled "Please Select". Below the back view is a button labeled "Lower Back X". To the left of the back view is a radio button labeled "No" under the heading "Missed Attempted Contact With Subject". At the bottom of the window are two buttons: "Done With Actions" and "+ Add More Actions".

FIG. 11

1200 ↘

Subject Action	Officer Action	Subject Action Score	Officer Action Score
Flee	Physical-Used Arm Bar	1	2
Subject-Physical-Head	CED	2	3
	Total	3	5
	Score		
Starting Subject Action	1		
Starting Officer Action	2		
Starting Proportionality	0		
Starting Escalation	1		
Total Subject Action	3		
Total Officer Action	5		
Subject Highest Action	2		
Officer Highest Action	3		
Highest Action Proportion	0		
Highest Proportionality	0		
Subject Escalation	1		
Officer Escalation	2		
Number of Interactions	2		

FIG. 12

Use of Force Activity					
	Outlier Status	Officer	Dept	Unit	Peer Group
Use of Force	Expected	10	10	10	10
Total Percent Adverse Events	Very High	6 (20%)	4 (10%)	4 (10%)	4 (10%)
Total Percent Disproportionate Force	High	3 (60%)	1 (10%)	1 (10%)	1 (10%)
Total Percent Subject Resistance	Expected	4 (10%)	4 (10%)	4 (10%)	4 (10%)
Avg Force Severity	Expected	34	34	34	34
Avg Uof / Arrests	Expected	1/12	1/13	1/13	1/13

FIG. 13

SYSTEMS AND METHODS FOR GENERATING USE OF FORCE INDICATORS

CROSS REFERENCE TO RELATED APPLICATIONS

[0001] This application is a continuation of U.S. patent application Ser. No. 17/370,870 filed Jul. 8, 2021, which claims the benefit of U.S. provisional patent application No. 63/049,654, filed Jul. 9, 2020.

FIELD OF THE INVENTION

[0002] This invention relates generally to a system and method for generating a score or scores for an incident involving a police officer's use of force in an incident. The system generates an overall score for an officer, for an incident and for each individual interaction within an incident. The scores can be compared to the individual or average scores for other officers or groups of officers to determine if a particular officer or an officer in a particular incident used more force than average or than expected in an incident. The scores can also be compared to policies of a department to determine if a particular incident or interaction within an incident was within a department's policy.

BACKGROUND OF THE INVENTION

[0003] Police officers encounter individuals on a daily basis. From time to time, incidents occur where an officer uses force during an interaction or in a series of interactions with on another individual or subject.

[0004] There is a need for system that can evaluate whether an officer's use of force in a particular incident was more or less than expected as compared to another group of officers or if a particular use of force interaction was not in policy for the department. There is also a need for a system that can provide scores for incidents that involve use of force to rank an officer's use of force with respect to other officers for similar incidents.

SUMMARY

[0005] These and other needs are addressed by the present invention, in which data or information about use of force incidents is collected, stored and processed by the system. The data can be analyzed and used to create and identify a model using machine-learning to create a metric from an officer's use of force scores. The metric can then be used in a machine-learning system to determine whether a particular officer is at risk of an adverse event.

[0006] Generally, the present system collects, stores and processes information for an incident involving a police officer's use of force. For example, police officer data and subject data can be entered and stored in the system. Also information about the incident can be entered and stored, such as location, date and time, weather conditions, etc. The system can also collect and store information about the type of force applied in the incident. Types of force may include, vehicle pursuit, physical action, OC spray, impact weapon, ECW discharge, or firearm usage. Other information that can be collected and stored about the incident to document the incident may include warnings, use-of-force sequence, charges, resisting arrest, injuries, etc. A detailed description of an incident may also be entered and stored, such as narrative reports from the police officer and any subject statements.

[0007] The present invention stems from the realization that in the context of use of force incidents, there is a need to identify whether an officer's actions and/or escalation of use of force are in line with policy or not and to determine whether an officer's use of force and escalation of use of force are greater than expected in a particular incident.

[0008] Accordingly, one aspect of the invention relates to a method and software for collecting and storing various data about incidents that involve use of force and determining scores for such uses of force.

[0009] Other advantages of the present invention will become readily apparent from the following detailed description. The invention is capable of other and different embodiments, and its several details are capable of modifications in various obvious respects, all without departing from the invention. Accordingly, the drawing and description are illustrative in nature, not restrictive.

BRIEF DESCRIPTION OF THE DRAWINGS

[0010] The foregoing aspects and many of the attendant advantages of the disclosed embodiments will become more readily appreciated by reference to the following detailed description, when taken in conjunction with the accompanying drawings, wherein:

[0011] FIG. 1 is a block diagram illustrative of an embodiment of a computing environment for generating use of force indicators for a police officer's use of force in an incident;

[0012] FIG. 2 is a flow diagram illustrating an example of a sequence of interactions in a first incident.

[0013] FIG. 3 is an illustration of an example user interface provided by the system for display by the user computing device.

[0014] FIG. 4 is an illustration of another example user interface provided by the system for display by the user computing device.

[0015] FIG. 5 is an illustration of another example user interface provided by the system for display by the user computing device.

[0016] FIG. 6 is an illustration of another example user interface provided by the system for display by the user computing device.

[0017] FIG. 7 is an example of a display showing use of force indicators generated by the system for a police officer's use of force in the first incident.

[0018] FIG. 8 is a flow diagram illustrating an example of a sequence of interactions in a second incident.

[0019] FIG. 9 is an illustration of another example user interface provided by the system for display by the user computing device.

[0020] FIG. 10 is an illustration of another example user interface provided by the system for display by the user computing device.

[0021] FIG. 11 is an illustration of another example user interface provided by the system for display by the user computing device.

[0022] FIG. 12 is an example of a display showing use of force indicators generated by the system for a police officer's use of force in the second incident.

[0023] FIG. 13 is an example of a display showing a profile diagram for an officer on use of force.

DETAILED DESCRIPTION

[0024] A system and methodology for determining a use of force score for an officer or incident and comparing that use of force score to use of force scores or average use of force scores, for other officers or groups of officers is described. In the following description, for the purposes of explanation, numerous specific details are set forth in order to provide a thorough understanding of the present invention. It will be apparent, however, to one skilled in the art that the present invention may be practiced without these specific details. In other instances, well-known structures and devices are shown in block diagram form in order to avoid unnecessarily obscuring the present invention.

[0025] Turning now turn to FIG. 1, a block diagram illustrating an embodiment of a computing environment **100** for a system for collecting, storing and processing information for one or more incidents involving a police officer's use of force is shown. The computing environment **100** includes a use of force generation system **102**, a user computing device **104**, and a data storage device or system **106**, each in communication via a network **110**.

[0026] Embodiments of the use of force generation system **102** and the user computing device **104** may be independently selected by any computing device such as desktop computers, laptop computers, mobile phones, tablet computers, server computers, client computers, and the like. Embodiments of the data storage device or system **106** may include one or more data storage devices capable of maintaining computer-readable data. Examples may include, but are not limited to, magnetic storage (e.g., hard disk drives, etc.), solid state storage (e.g., flash memory, etc.), network storage, and other computer-readable media known in the art. Embodiments of the network **110** may include, but are not limited to, local area networks (LANs), wide area networks (WAN), the Internet, wired networks, wireless networks), and telephone networks.

[0027] The use of force system **102** includes a one or more components, shown generically as **103A**, **103B** and **103C**, used to collect, store and process various types of information and to generate various use of force indicators and scores. As described in more detail below, various components can be used to store or generate scores for actions, proportionality scores and escalation scores. Further, the various components may store or generate this information for an individual action, an interaction with several actions, for an officer and for a group of officers. One or more components may be used to analyze use of force data and to create and identify a model using machine-learning to create a metric from an officer's use of force scores. One or more components may use machine-learning to determine whether a particular officer is at risk of an adverse event. Furthermore, while the system **102** is illustrated in FIG. 1 as a single device, it may be understood that functionalities of one or more of the components **103A-C**, may be performed by a distributed computing environment including a plurality of computing devices, without limit.

[0028] Similarly, data storage device or system **106** may store one or more types of data to be used by the use of force system **102**. For example, and without limitation, general information about the incident **107A**, type of use of force used **107B**, officer **107C**, subject **107D** and interaction **107E**, can be collected and stored in the system **106**.

[0029] General information about the incident **107A** may include location information, date and time information and

weather conditions. General information about the incident may also include the case number, the names of the officers and subjects as well as the age, race, address and other pertinent information about the officer and subject.

[0030] Information about the type of force used or applied in the incident **107B** can also be collected and stored in the system. The specific type of force used or applied may be identified, such as vehicle pursuit, physical action, OC spray, impact weapon, ECW discharge or firearm usage. Other information that can be collected and stored about the incident may include warnings, use-of-force sequence, charges, resisting arrest, injuries, etc. associated with the incident. A detailed narrative description of an incident may also be entered and stored, such as narrative reports from the police officer and any subject statements.

[0031] Officer information **107C** can also be collected and stored for an incident. Officer information can include the officer identification information (e.g., officer name, officer number, etc.) as well as information about the officer at the time of the incident, e.g., whether the officer was in uniform or not, whether the officer was in a marked or unmarked police car, whether the officer was on duty, whether the officer was alone or with a partner, etc. If there was another responding officer, similar information for that other officer can be entered and stored for the incident as well.

[0032] Subject information **107D** can also be collected and stored for an incident. For example, subject information can include a subject's date of birth, occupation, whether the subject was armed, etc.

[0033] Information about the interaction for an incident **107E** can also be collected and stored. Interaction information may include information about a single interaction or a series of interactions within an incident. Generally, the interaction information provides information about who contacted who, what type of contact was made, where on a person's body the contact took place and whether the subject resisted. For example, for a taser incident, a first interaction may be that the subject hit an officer with his fist in the officer's left arm. A second interaction may be that the officer responded by discharging a taser at the subject's stomach.

[0034] Also, if a subject resisted an officer's request, that information can be collected as well. Similarly, if there is more than one interaction for an incident, information about each interaction can be collected and stored as well.

[0035] In addition, any injury information resulting from the incident can also be collected and stored. For example, if the officer or subject was injured as part of any of incidents or the interaction, that information can be collected and stored in the system.

[0036] Information about any weapon (e.g., a taser) used in an incident or interaction may also be collected and stored in the system. For example, name serial number and manufacturer of the taser may be collected.

[0037] The system can analyze the use of force for an incident or a single or several interactions based on the information collected and stored about the incident and other incidents over time.

[0038] FIG. 2 illustrates a schematic flow diagram of a sequence or interactions in a first example of an incident **200**. The incident **200** shows the following sequence of interactions relating to a traffic stop **202**:

[0039] Interaction 1 (**204**): An officer asked a subject to get out of their car.

[0040] Interaction 2 (206): The subject refused and spit at an officer.

[0041] Interaction 3 (208): The officer pulled the subject through the window of his car

[0042] Interaction 4 (210): The subject resisted and punched the officer in the head

[0043] Interaction 5 (212): The officer used chemical spray in the subject's face and restrained the subject.

[0044] In the above sequence of interactions, the types of use of force applied by the officer were: asking the subject to get out of the car (Interaction 1) (204), pulling the subject through the window (Interaction 3) (208) and using chemical spray (Interaction 5) (212). Interactions 3 and 5 (208 and 212) are escalations of use of force by the officer in response to interactions from the subject as the use of force in Interactions 3 and 5 were greater than in the previous subject interactions.

[0045] As shown in FIG. 3, a user interface 300 with pre-populated drop down menus for interactions types and names, etc., can be provided by one or more components 103A-C in the system 102 to enter the information from the incident into the system 102 in an easy and more uniform manner and to store the information in the data storage system 106. As shown in FIG. 3, the information that can be entered may include information about the initial contact in block 302, including the reason for the initial contact. The information may be entered also may include information about the initial interaction and subsequent interaction(s) in blocks 304 and 306.

[0046] As shown in block 302, the reason for the initial contact can be entered via a drop down (e.g., a traffic stop) 302A.

[0047] As shown in block 304 information about the first interaction that can be provided via one or more drop downs is an identification of the person initiating the interaction (e.g., Officer Smith) 304A, the type of interaction (e.g., use of verbal command) 304B, and an identification of the person who the interaction was with (e.g., subject John Doe) 304C. Other information can be included via drop down, such as the position of the subject (e.g., the subject was sitting) 304D. Note that the officer's name can be used in these drop downs (as shown in 304A) in order to enable the system to compile information in the database or storage system 106 about the officer concerning this incident, as well as other incidents involving the same officer.

[0048] Similarly, as shown in block 306, information relating to a response to the first interaction can be provided via drop downs. For example, here, an identification of the person initiating the next interaction (e.g., subject John Doe) 306A, the type of interaction (e.g., spit at) 306B, and an identification of the person who the interaction was with (e.g., Officer Smith) 306C. Other information 306D can be included via drop down, such as the position of the subject (e.g., the subject was in the car) 306E.

[0049] Also as shown in FIGS. 4-6, in addition to drop downs, other interfaces can be provided by the system 102 to enter information about an interaction, including diagrams of a person's body to allow a user to indicate by selection (such as with a mouse, track pad or touch screen) what part of the body was contacted during certain types of interactions and toggles for yes or no answers to questions.

[0050] As shown in FIG. 4, a different user interface 400 may be used to enter information about the next interaction. For example, drop downs are used to enter information that

Officer Smith 402, used a take down 404 on the subject 406. Then, using diagrams 408, a user can indicate by indicating in the diagram where the force was applied to the subject. As shown, the force was applied to the subject's right and left chest 410 and 412 and right and left shoulder 414 and 416. As shown, when an area is indicated or selected, it becomes highlighted to provide a visual confirmation of the selected area. The information entered by touching the diagram may also appear as text in a window 418 as another confirmation. Other information can be entered by drop down such as the position of the officer (e.g., standing) 420. Other information requested as "yes" or "no" answers, can be entered by toggle input, such as whether a protective barrier was used and whether the subject was struck (422).

[0051] As shown in FIG. 5, another user interface 500 may be used to enter information about the next interaction 500. Again, drop downs are used to enter information that the subject 502, used fists 504 on Officer Smith 506. Then, using diagrams 508, a user can indicate that the fists contacted the officer's right cheek 510, e.g., by touching the area on a diagram on an input screen. The information entered by touching the diagram may also appear as text in a window 512. Other information can be entered by drop down such as the position of the subject when applying the action (e.g., in the car) 514. Other information requested as "yes" or "no" answers, can be entered by toggle input, such as whether a protective barrier was used and whether the subject was struck 422.

[0052] As shown in FIG. 6, another user interface 500 may be used to enter information about the next interaction 600. Again, drop downs are used to enter information that Officer Smith 602, discharged chemicals 604, at the subject 606. Then, using diagrams 608, a user can indicate by touching the diagram that the chemicals were discharged in the subject's right eye, left eye and nose 610, 612 and 614. The information entered by touching the diagram may also appear as text in a window 616. Other information can be entered by adding text into boxes 618, 620, 622 and 624 such as the chemical type and chemical serial number, as well as the distance when the chemicals were first used and the distance when the chemical were last used. Other information can be added by additional drop downs 626 such as the position of the subject when applying the action. Other information requested as "yes" or "no" answers, can be entered by toggle inputs 628, 630, 632 and 634, such as whether a protective barrier was used, whether the subject was struck, whether the subject was decontaminated after usage and whether the chemical discharge was accidental.

[0053] With reference to the incident shown in FIGS. 2 and 3-6, FIG. 7 illustrates a screen 700 showing various scores that can be generated and/or displayed by one or more components 103A-C of the system 102 relating to the incident. For example, a score can be determined for each of the actions in the incident. In this example, the subject's "traffic violation" action is assigned a score of "0", and the officer's "verbal command" action is assigned a score of "1". In addition, the subject's "spit" action is assigned a score of "1", and the officer's "physical takedown" action is assigned a score of "2". In addition, the subject's "punch" action is assigned a score of "3", and the officer's "chemical spray" action is assigned a score of "3"

[0054] Other scores can be displayed or generated for the incident, such as scores for the "Starting Subject Action" (here, the "Traffic Violation"), "Starting Officer Action"

(here, the “Verbal Commands”, “Starting Proportionality”, “Starting Escalation” (here, the difference between the scores for “Verbal Command” and “Traffic Violation”), Total Subject Action (here, the sum of the scores for the three subject actions), Total Officer Action (here, the sum of the scores for the three officer actions), “Subject Highest Action” (here, the highest action score is “2” for punch); “Officer Highest Action” (here, the highest action score is “3” for chemical spray), “Highest Action Proportion”, “Highest Proportionality”, “Subject Escalation” and “Officer Escalation”).

[0055] The other values listed above are defined as follows:

[0056] “Starting Proportionality”: The proportionality of the first application of force that the officer uses in response to resistance.

[0057] “Highest Action Proportionality”: The proportionality of the highest resistance received and the highest force employed (regardless of where in the sequence they occurred).

[0058] “Highest Proportionality”: The highest proportionality score found between each resistance/force step, regardless of where in the sequence it occurred. For example, if the first action was proportional with a score of 0, then the second action was highly disproportionate (e.g. using a firearm on passive resistance received with a score of 4), then the subject resistance increases and the officer force decreases (proportionality score might return to 0).

[0059] “Subject Escalation”: The amount of change in resistance received from the subject in relation to the force used by the officer. (e.g., initially passive resistance then increases to flee, then increases to assault are all escalations in force all while the officer uses verbal commands would be escalation in resistance from the subject). An escalation at each step of 1 relative to the officer force and the previous resistance the subject used.

[0060] “Officer Escalation”: The amount of change in force employed by the officer relative to their previous force and the escalation in resistance received from the subject. For example, where the subject continues to passively resist and the officer continues to escalate to higher and higher levels of force severity.

[0061] An overall score can also be determined by the system **102** for the overall interaction, i.e., the entire sequence of events or interactions in the incident. The overall score can be used to determine whether the overall force used by the officer was proportional to the resistance encountered from the subject. These scores can be measures of severity of an action and the proportionality of an action. The severity of an action is a measure of the level of the action itself (e.g., a measure of the level of an officer’s use of force or a measure of the level of a subject’s resistance), whereas the measure of the proportionality of an action is a measure of the action’s level compared to a previous actions level, e.g., the measure of the level of an officer’s use of force compared to the level of a subject’s resistance. Each of these measures can be computed as averages or as overall scores. They can also be compared to aggregate scores or to scores for an officer’s peers, i.e., a certain group of officers.

[0062] The system **102** can also determine for each step in the incident whether the force applied by the officer in that step was proportional to the resistance by the subject.

[0063] The system **102** can also look across each step to determine whether a change in the level of resistance by the subject resulted in a proportional change in use of force by the officer. For example, if the subject’s resistance increased by one level, the scores may reflect whether the officer’s use of force increased to a level that is one level above the resistance level (which may be within policy) or whether it increased by more than one level (which may not be within policy).

[0064] By determining the scores for the use of force and for the resistance levels, the system **102** can determine average escalation and average proportionality scores for a variety of different metrics, e.g., for an officer, for similar incidents, or for departments.

[0065] The goal is to determine whether the use of force is more or less that should be expected for a particular incident.

[0066] The system **102** can be used to process the information to determine: (1) whether the use of force by the officer in each interaction/step was proportional; (2) whether the overall result of the escalation was greater than expected and (3) provide a baseline of the severity that should have been used.

[0067] The system **102** can also generate a profile for an officer and/or for all officers or for a group of officers (e.g., all officers within a particular department or all officers within a particular location). The system **102** can determine an average score for use of force for an officer and compare it to an overall average use of force for a particular group of officers. For example, an officer’s score can be compared to others, as follows: a composite peer group average is calculated for one or more metrics. The officer is then judged to be high/very high based on how far away the officer is from the peer group average. For example, an officer one standard deviation above the peer group average is considered high. The peer group model (and the risk profile model) may be determined by a system such as one described in U.S. patent application Ser. No. 17/132,458 filed Dec. 23, 2020, the specification of which is incorporated herein by reference. An example of a display screen showing a profile diagram on use of force is illustrated in FIG. 13. The disproportion force counts and percentages are shown in the rows for “Total|Percent Adverse Events” and “Total|Percent Disproportionate Force” which show Outlier Status of “Very High” and “High”, respectively. The display for these outliers may be color coded relative to peer group. The remaining rows show use of force counts and percentages that are “Expected”.

[0068] The officer’s profile may also be used as a metric that can be input into a machine learning system to help determine whether a particular officer is at risk of a potential adverse event, such as a system described in U.S. patent application Ser. No. 17/132,458, the specification of which is incorporated herein by reference. The machine learning system may also be a component **102A-C** of the system **102**.

[0069] An officer’s profile based on use of force can be generated and displayed visually on a user interface that illustrates the officer’s profile alone and/or compared to a particular group of officers.

[0070] FIG. 8 illustrates a schematic flow diagram of a sequence or interactions in a second example of an incident **800**. The incident shows the following sequence of interactions relating to a subject that is fleeing **802**:

[0071] Interaction 1 (804): An officer used an arm bar to control the subject.

[0072] Interaction 2 (806): The subject responded by hitting the officer in the nose with his head.

[0073] Interaction 3 (808): The officer escalates by using electronic discharge weapon on the subject.

[0074] In the above sequence of interactions, the types of use of force applied by the officer were: using an arm bar to control the subject Interaction 1 (804) and using an electronic discharge weapon on the subject Interaction 3 (808). Interaction 3 (808) is an escalation of use of force by the officer in response to the interaction from the subject. After Interaction 3, the use of force incident is complete 810

[0075] Although not shown for this incident, as described above, pre-populated drop down menus with interactions types and names, etc., can be provided to enter the information from the incident into the system 102 in an easy and more uniform manner and to store the information in the data storage system 106.

[0076] Also as shown in FIGS. 9-11, various user interfaces 900, 1000 and 1100 with diagrams of a person's body and various drop downs can be provided by the system to allow a user to indicate by selection what part of the body was contacted and to enter other information about each interaction in an incident. These interfaces and methods of selection are similar to those described above with respect to FIGS. 4-6.

[0077] In FIG. 9, the user interface 900 provides drop downs to enter information that Officer Smith 902, used an arm bar 904 on the subject 906. Then, using diagrams 908, a user can indicate by touching the diagram on the screen that the arm bar was applied to the subject on the subject's right arm 910.

[0078] In FIG. 10, the user interface 1000 provides drop downs to enter information that the subject 1002, used his head 1004 on the officer 1006. Then, using diagrams 1008, a user can indicate by touching the diagram that the head was used applied to the officer's nose 1010. The information entered by touching the diagram may also appear as text in a window 1012. Other information requested as "yes" or "no" answers, can be entered by toggle input, such as whether the officer missed attempted contact with the subject (1014).

[0079] In FIG. 11, the user interface 1100 provides drop downs to enter information that the officer 1102, used an electronic discharge weapon ("CED") 1104 on the subject 1106. Then, using diagrams 1108, a user can indicate by touching the diagram on the screen that the weapon was applied to the subject's lower back 1110. The information entered by touching the diagram may also appear as text in a window 1112. Other information requested as "yes" or "no" answers, can be entered by toggle input, such as whether the officer missed attempted contact with the subject 1114.

[0080] With reference to the incident shown in FIG. 8, FIG. 12 illustrates a screen 1200 various scores that can be generated for the incident. For example, a score can be determined for each of the interactions. In this example, the subject's "flee" action is assigned a score of "0", and the officer's "physical arm bar" action is assigned a score of "2". In addition, the subject's "physical head to the officer's nose" action is assigned a score of "2", and the officer's use of an "electronic discharge weapon" is assigned a score of "3".

[0081] Other scores can be displayed or generated for the incident, such as scores for the "Starting Subject Action" (here, "fleeing"), "Starting Officer Action" (here, the "arm bar"), "Starting Proportionality", "Starting Escalation" (here, the difference between the scores for "fleeing" and "arm bar"), Total Subject Action (here, the sum of the scores for the two subject actions), Total Officer Action (here, the sum of the scores for the two officer actions), "Subject Highest Action" (here, the highest action score is "2" for use of head); "Officer Highest Action" (here, the highest action score is "3" for use of an electronic discharge weapon), "Highest Action Proportion", "Highest Proportionality", "Subject Escalation" and "Officer Escalation". The other values listed here are described above. This illustration also shows a total number of interactions.

[0082] Accordingly, a system, method, software, and variables for evaluating a police officer's use of force are described. More specifically, techniques are disclosed, wherein a system can construct collect, store and process information that can be used to determine use of force scores for an officer and to compare that officer's score to average scores for another individual or group of officers or to determine whether an officer's use of force in a particular incident it within policy or not.

1-13. (canceled)

14. A computer-implemented method for obtaining and storing interaction information that may be used to determine a score reflecting the use of force that a police officer applied in an incident involving use of force, comprising:

providing a computer system having a memory and a display;

collecting and storing in the computer system's memory, information about the police officer involved in the incident;

collecting and storing in the computer system's memory, information about a subject involved in the incident;

displaying a user interface on the display, the user interface containing:

interaction information about a sequence of interactions in the incident, the interaction information including for each interaction of the sequence of interactions:

information about an identification of a person initiating an interaction,

information about a type of interaction, and

information about an identification of a person who the interaction was with, and

interaction information about where on a human body force was applied to, including a diagram of the human body,

upon a user's selection of interaction information, storing the interaction information in the computer system's memory.

15. The computer-implemented system of claim 14, wherein:

the interaction information about where on a human body force was applied to, is touch input from a user by touching a location on the diagram of the human body.

16. The computer-implemented system of claim 15, wherein:

the interaction information about where on a human body force was applied to, is touch input from a user by

touching a location on the diagram of the human body and is a single location on the diagram of the human body.

17. The computer-implemented system of claim **15**, wherein:

the interaction information about where on a human body force was applied to, is touch input from a user by touching a location on the diagram of the human body and includes two or more locations on the diagram of the human body.

18. The computer-implemented system of claim **14**, wherein:

in response to collecting and storing interaction information selected about where on a human body force was applied to, the user interface displays text information about the selected area of the body.

19. The computer-implemented system of claim **14**, wherein:

in response to collecting and storing interaction information selected about where on a human body force was applied to, the user interface displays a visual confirmation of the selected area of the body.

20. A computer system for collecting, storing and processing data relating to an incident involving the police officer's use of force during an incident, comprising:

- a component with instructions stored in memory and executable by a processor for collecting information about the police officer involved in the incident;
- a component with instructions stored in memory and executable by a processor for collecting information about a subject involved in the incident;

a display with a user interface for displaying information about one or more interactions in the incident to be selected by a user, including:

- information about an identification of a person initiating an interaction,
- information about a type of interaction, and
- information about an identification of a person who the interaction was with, and

a diagram of the human body with information about a location as to where force was applied to the subject;

a component coupled to the display for sending information selected about the location as to where force was applied to the subject, to the component for collecting information about a subject involved in the incident.

21. The computer system of claim **20**, wherein:

the display is a touch screen that allows a user to make a selection on the diagram of the human body as to where force was applied to the subject by touching a location on the diagram of the human body where the force was applied.

22. The computer system of claim **21**, wherein:

a component that, in response to receipt of the user's selection from the diagram of the human body, displays on the display, text information about the selected area of the body.

23. The computer system of claim **21**, wherein:

a component that, in response to receipt of the user's selection from the diagram of the human body, displays on the display, a visual confirmation of the selected area.

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